

Definition: An **R-mixed Hodge structure** on  $V$  consists of two filtrations:

- a weight filtration  $W_\bullet$  on  $V \otimes \mathbb{Q}$  (increasing),
- a Hodge filtration  $F^\bullet$  on  $V_{\mathbb{C}} = V \otimes \mathbb{C}$  (decreasing),

$$\begin{aligned} \dots &\leq W_{k-1} \leq W_k \leq W_{k+1} \leq \dots \\ \dots &\supseteq F^{k+1} \supseteq F^k \supseteq F^{k-1} \supseteq \dots \end{aligned}$$

such that the induced filtration of  $F^\bullet$  on

$$\mathrm{Gr}_k^W(V \otimes \mathbb{Q}) = W_k / W_{k-1}$$

is a Hodge filtration of weight  $k$ .

Its Hodge numbers are

$$h^{p,q}(V) = \dim_{\mathbb{C}} \mathrm{Gr}_F^p \mathrm{Gr}_{p+q}^W(V_{\mathbb{C}}).$$

These are the coefficients of the Hodge-Euler polynomial

$$e_{\mathrm{Hdg}}(V) = \sum_{p,q \in \mathbb{Z}} h^{p,q}(V) u^p v^q \in \mathbb{Z}[u, v, u^{-1}, v^{-1}]$$

Def: Let  $(H_{\mathbb{Z}}, F, W)$  be a MHS. We define its **Tate twist** to be the MHS

$$H(m)_{\mathbb{Z}} = (2\pi i)^{-m} H_{\mathbb{Z}},$$

$$W_k(H(m)) = W_{k+m}(H),$$

$$F^p(H(m)) = F^{p+m}(H).$$

A morphism  $f: (V, F) \rightarrow (V', F')$  of filtered vector spaces induces linear maps  $\text{Gr}_F^p V \rightarrow \text{Gr}_{F'}^p V'$ . This is not nec. injective, even if  $f$  is.

We say that  $f$  is **strict** if

$$f(V) \cap F^p V' = f(F^p V) \quad \forall p.$$

In this case,  $\text{Gr}_F^p f$  is injective if  $f$  is.

Example: A morphism of (pure) Hodge structures is strict. This follows from the equivalence of Hodge filtrations and Hodge decompositions.

We want to show that morphisms of mixed Hodge structures are strict. We do this by introducing a splitting similar to the Hodge decomposition.

Definition: The **Deligne splitting** of a real MHS  $V$  is the bigrading

$$V_{\mathbb{C}} = \bigoplus I^{p,q},$$

where

$$I^{p,q} = F^p \cap W_{p+q} \cap (\overline{F}^q \cap W_{p+q} + \sum_{j \geq 2} \overline{F}^{q-j+1} \cap W_{p+q-j}).$$

Note that

$$\bullet W_k^{\mathbb{C}} := W_k \otimes \mathbb{C} = \bigoplus_{p+q \leq k} I^{p,q},$$

$$\bullet F^p = \bigoplus_{r \geq p} I^{r,s},$$

For a proof, see  
Lemma-Definition 3.4

$$\bullet I^{p,q} = \overline{I}^{q,p} \mod W_{p+q-1}^{\mathbb{C}}.$$

In other words, the bigrading is a **weak splitting** of the real MHS.

Similar to the pure case, it follows that morphisms of MHS are strict for both filtrations. In particular, morphisms of MHS which are isomorphisms on  $\mathbb{R}$ -modules are isomorphisms of MHS.

It also follows that taking  $\text{Gr}_k^W, \text{Gr}_F^p, \text{Gr}_F^p \text{Gr}_k^W$  are exact.

Finally, it follows that the map  $H^*/\ker \rightarrow \text{Im} \phi$  is an isomorphism of MHS and that the category of MHS is abelian.

We want to give the cohomology of all complex algebraic varieties a MHS. We start with smooth varieties.

Let  $U$  be a smooth variety. Then  $U$  can be embedded <sup>Zariski openly</sup> into a compact variety  $X$  such that  $D = X \setminus U$  is a **normal crossing divisor** (looks like the crossing of coordinate hyperplanes). We say that  $D$  is **simple**, or **strict**, if its irreducible components are smooth. In this case,  $X$  is a **good compactification**.

A holomorphic differential form  $w$  on  $U$  has **logarithmic poles along  $D$**  if  $w, \partial w$  have at most simple poles along  $D$ . They form the **logarithmic de Rham complex**  $\Omega_X^*(\log D) \xleftarrow{j_*: U \hookrightarrow X} \Omega_U^*$ .

If  $X$  is a good compactification,  $(z_1, \dots, z_n)$  coordinates in which  $D = \{z_1, \dots, z_k = 0\}$ , then

$$\Omega_X^1(\log D)_p = \bigoplus_{X,p} \frac{\partial z_1}{z_1} \oplus \dots \oplus \frac{\partial z_k}{z_k} \oplus \bigoplus_{X,p} \partial z_{k+1} \oplus \dots \oplus \bigoplus_{X,p} \partial z_n,$$

$$\Omega_X^p(\log D)_p = \wedge^p \Omega_X^1(\log D)_p.$$

Lemma:  $H^k(U; \mathbb{C}) = H^k(X, \Omega_X^*(\log D))$ . Actually,  $\Omega_X^*(\log D) \xrightarrow{j_*} \Omega_U^*$  is a quasi-isomorphism.

For the main theorem, define the filtration

$$W_m \Omega_X^p(\log D) = \begin{cases} 0 & \text{for } m < 0, \\ \Omega_X^p(\log D) & \text{for } m \geq p, \\ \Omega_X^{p-m} \wedge \Omega_X^m(\log D) & \text{for } 0 \leq m \leq p. \end{cases}$$

This induces a filtration

$$W_m H^k(U; \mathbb{C}) = \varinjlim (H^k(X, W_{m-n} \Omega_X^n(\log D))) \longrightarrow H^k(U; \mathbb{C}).$$

Similarly, the trivial filtration  $F$  on  $\Omega_X^*(\log D)$  gives

$$F^p H^k(U; \mathbb{C}) = \varinjlim (H^k(X, F^p \Omega_X^*(\log D))) \longrightarrow H^k(U; \mathbb{C})$$

Theorem: The filtration  $W_m H^k(U; \mathbb{C})$  can be defined over  $\mathbb{Q}$ . Furthermore,  $(H^k(U), W, F)$  is a MHS.

*This is also true for singular varieties*